

Exact and Floating-Point Numbers

- 1 Exact and Floating-Point Numbers
 - delayed evaluation
- 2 Rational Approximations
 - approximations with a prescribed accuracy
 - continued fractions
- 3 Interval Arithmetic
 - functions of intervals

MCS 320 Lecture 4
Introduction to Symbolic Computation
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Exact and Floating-Point Numbers

In symbolic computation we compute with symbols and expressions.

- 1 This implies that numerical approximations are delayed.
- 2 Rational approximations allow for exact computations.

In the context of working with approximations for π , consider the difference between symbolic and numerical computing.

- 1 Computing numerically, work with an expansion, e.g.: 3.14.
- 2 Computing symbolically, use a rational approximation, e.g.: $22/7$.

In numerical analysis, we worry about error propagation, in symbolic computation, expression swell is a problem.

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Approximations with a Prescribed Accuracy

To compute a rational approximation with a prescribed accuracy, execute the following two steps:

- 1 Evaluate a symbol in a number system.
- 2 Construct a rational approximation for the evaluation.

Example:

- 1 Evaluate π with five decimal places: 3.1416.
- 2 The rational approximation is then 355/113.

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Continued Fractions

Definition (continued fraction)

A *continued fraction* is

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots + \frac{1}{a_n}}}}.$$

defined by a list of natural numbers $[a_0; a_1, a_2, \dots, a_n]$.

The rational numbers defined by the sublists are called *the convergents* of the continued fraction.

Example: The continued fraction of $\sqrt{2}$ is defined by $[1; 2, 2, 2, 2, \dots]$.

$$[1; 2, 2, 2, 2] \text{ defines } 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}}} = \frac{41}{29}$$

Consecutive Rational Approximations

The convergents of $[1; 2, 2, 2, 2]$ are $[1, 3/2, 7/5, 17/12, 41/29]$.

The first convergents for π give the consecutive rational approximations

$$3, \frac{22}{7}, \frac{333}{106}, \frac{355}{113}, \frac{103993}{33102}, \frac{104348}{33215}, \frac{208341}{66317}, \frac{312689}{99532}.$$

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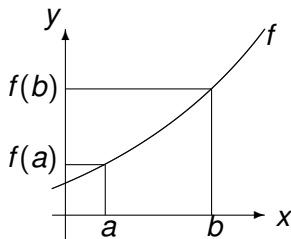
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functions of intervals



For continuous f over $[a, b]$: $f([a, b]) = \left[\min_{x \in [a, b]} f(x), \max_{x \in [a, b]} f(x) \right]$.

- Let $f \in \{+, -, \times, /\}$ and we have *interval arithmetic*.
- The inverse: $1/[a, b]$ is well defined if $0 \notin [a, b]$.

Interval arithmetic allows to keep track of errors in a computation.